Won Jun Lee

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EDUCATION

Purdue University **Aug 2021 – May 2027 (expected)**
Ph.D. Candidate, Electrical and Computer Engineering ***(GPA: 3.81/4.0)*** *West Lafayette, IN, U.S.*
Advisor: Prof. Chih-Chun Wang

Yonsei University **Mar 2015 – Feb 2021**
B.S., Electrical and Electronics Engineering ***(GPA: 3.84/4.0)*** *Seoul, South Korea*

RESEARCH EXPERIENCE

Prof. Chih-Chun Wang's Lab **Aug 2021 – Present**
Research Assistant *Purdue University, West Lafayette, IN, U.S.*

- **AoI-Optimal Scheduling and Resource Allocation for Multi-Channel Wireless Systems [C2, J1]**
  - Characterized the **AoI-optimal scheduling policy** for arbitrary K-channel systems as a **semi-Markov decision process** on the age state: threshold (water-filling) structure with closed-form solutions, and **reduced policy computation from** $O(N^3)$ **to** $O(N)$.
  - Extended the framework to a **joint latency–cost–quality–freshness optimization** over heterogeneous channels (e.g., cloud vs. edge server selection), with a **structured channel-selection algorithm** that scales to large systems.
- **Cost-Aware Multi-Sensor Scheduling for Remote Estimation under Correlated Noise [C3]**
  - Developed a **cost-aware sensor-scheduling framework** that decides which sensors transmit under communication and sensing budgets, trading estimation error against cost.
  - Designed an estimation method that **alleviates the computational burden of Kalman filtering** under **correlated noise** and remains tractable for high-dimensional systems.

Digital Communications Lab **Jan – Jun 2020**
Undergraduate Research Intern *Yonsei University, South Korea*

- **5G Network Performance Benchmarking in Heterogeneous Environments**
  - Built an Android network-analysis application (**UDP socket programming**) to measure end-to-end latency and throughput in heterogeneous 5G/Wi-Fi environments.
  - Implemented **UDP hole punching** for real-world connectivity benchmarking.

High Dimensional Signal Processing Lab **Jul – Aug 2019**
Undergraduate Research Intern *Yonsei University, South Korea*

- **DNN Classification Logic Analysis [C1]**
  - Assisted in analyzing decision boundaries of deep neural network classifiers.

PUBLICATIONS

[J1] W. J. Lee and C.-C. Wang, "Latency-, cost-, and freshness-aware scheduling over heterogeneous multi-channel systems," journal article, in preparation.

[C3] W. J. Lee and C.-C. Wang, "Cost-aware sensor scheduling for multi-sensor remote estimation under correlated noise," submitted to *IEEE Int. Conf. on Communications (ICC)*, 2027.

[C2] W. J. Lee and C.-C. Wang, "AoI-optimal scheduling for arbitrary K-channel update-through-queue systems," *in Proc. IEEE Int. Symp. on Information Theory (ISIT)*, 2024, pp. 957–962.

[C1] J. Kim, S. Woo, **W. J. Lee**, D. Kim, and C. Lee, "Analyzing Decision Polygons of DNN-based Classification Methods," *in Proc. 17th Int. Conf. Informatics in Control, Automation and Robotics (ICINCO)*, 2020, pp. 112–119.

PROFESSIONAL SERVICE

Reviewer, IEEE *International Symposium on Information Theory* (ISIT) **2026**

TECHNICAL SKILLS

Wireless: 5G/6G Resource Allocation, MAC Scheduling, Age of Information

Modeling & Optimization: MDPs, Stochastic Control, Convex Optimization, Probability, Information Theory, Reinforcement Learning (fundamentals)

Programming: MATLAB, Python (NumPy, PyTorch), C, Java

HONORS AND SCHOLARSHIPS

Ross Fellowship, Purdue University **2021 – 2025**

Student Travel Grant, IEEE International Symposium on Information Theory (ISIT) **2024**

Dean's List and Honors Scholarship, Yonsei University **2015–2016, 2019–2020**

MILITARY EXPERIENCE

KATUSA (Korean Augmentation To the United States Army) **Mar 2017 – Dec 2018**

Sergeant *Seoul, South Korea*

- Served as Information Management Officer (MOS 25B, IT Specialist) at the USFK Public Affairs Office, supporting mission-critical IT and communications operations.
- Led 6 soldiers as a **squad leader**, managing ~100 networked workstations and coordinating ROK–U.S. technical operations during **international joint military exercises** (Ulchi-Freedom Guardian 2017 / Foal Eagle 2018).